- Decisions are made by the distributed participants, not by any central authority.
- Intermediation is achieved by the use of automated software such as a blockchain.

Let’s explore the architectural elements of blockchain that address the unique needs of a decentralized system.

BLOCKCHAIN NODES, NETWORKS, AND APPLICATIONS

Consider air traffic. Flights have origins and destinations, and stopover airports and waypoints form the airline networks. Similarly, blockchain nodes host the computational environment that serves as endpoints of transactions and also performs other functions, such as relaying and broadcasting transactions.

> **DEFINITION** *Node* is a collective name for blockchain software and the machine or hardware on which it is installed for the participant of a decentralized system.

Figure 1.5 shows the logical architecture of a single blockchain node. A node can support many accounts to represent the identities of peer participants in the decentralized network. A 256-bit number represents an account. Compare this size with your traditional computer’s address size of 64 bits!